



An interpretable CNN framework for multi-class brain tumor classification with visual explanation in MRI scans

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ABSTRACT

Brain tumors are among the most life-threatening neurological conditions, and effective therapy and patient outcomes depend on accurate and timely diagnosis. This study proposes an interpretable and deployment-oriented deep learning framework for multi-class classification of brain tumors using magnetic resonance imaging (MRI). The framework employs a carefully designed lightweight CNN configuration that prioritizes computational efficiency and robustness for practical clinical use. Evaluations on a publicly available MRI dataset demonstrated the effectiveness of the proposed framework, achieving a classification accuracy of 99.54%. To enhance clinical trust, Grad-CAM visualizations were integrated to highlight class-discriminative regions within MRI scans, providing transparent and interpretable predictions. In addition, the framework was deployed within a user-friendly desktop application, enabling real-time diagnostic support. Comparative analysis with state-of-the-art models shows that the proposed system achieves competitive or superior performance while maintaining efficiency, making it a promising candidate for clinical decision support, subject to further validation in real-world settings.

1. Introduction

Brain tumors, which are characterized as a collection of abnormal cells that develop within the brain's tissues, are one of the main causes of illness and death globally. The World Health Organization (WHO) reports that brain tumors rank as the second most common cause of death globally [1]. Brain tumors are generally classified into two broad categories: benign and malignant. Benign tumors typically develop more slowly and neither invade nor spread to neighboring tissue. Malignant tumors, such as gliomas, on the other hand, have the ability to spread into nearby tissues and organs, which can lead to serious health problems. Early detection of brain tumors is crucial to improve patient survival rates and outcomes [2,3].

Magnetic Resonance Imaging (MRI) is the primary imaging modality for brain tumor detection due to its superior ability to visualize soft tissue structures. However, manual interpretation of MRI scans is still highly dependent on radiologists' expertise and is prone to human error, fatigue, and variability in judgment. Furthermore, brain tumors often exhibit heterogeneous appearances in MR images, with overlapping morphological features that complicate accurate classification. Biopsy procedures, although definitive, can be invasive and risky, especially during the early stages of diagnosis [4].

The application of deep learning techniques in medical imaging analysis has enabled automated detection and classification of brain tumors with high accuracy [5–7]. Convolutional neural networks (CNNs) have demonstrated strong capability in learning complex features from medical images and remain a dominant choice for image recognition tasks [8]. Despite this progress, several barriers still hinder clinical uptake. From a computer science perspective, explainability is essential for understanding model behavior, diagnosing failure modes, and improving robustness in safety-critical systems such as medical image analysis. From a medical perspective, interpretable AI predictions are necessary to support clinical trust, facilitate expert validation, and align automated decisions with established diagnostic reasoning. Consequently, explainability is not merely an optional visualization component but a fundamental requirement for advancing toward trustworthy clinical applications.

Many existing approaches focus primarily on improving classification accuracy, often at the expense of computational efficiency, interpretability depth, or practical deployment considerations. As a result, translating high-performing models into reliable, trustworthy, and clinically usable decision-support systems remains a challenge, particularly in resource-constrained settings.

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This study introduces an interpretable and deployment-oriented deep learning framework for multi-class brain tumor classification using MRI scans. While several existing approaches focus on architectural complexity or transfer learning, the proposed framework emphasizes a balanced design combining accuracy, interpretability, and practical usability. The key contributions of this work are summarized as follows:

- A carefully designed lightweight CNN configuration for four-class brain tumor classification, achieving high diagnostic accuracy while maintaining low computational complexity compared to transfer learning and transformer-based approaches.
- A multi-level explainability strategy using Grad-CAM, including layer-wise visualization and analysis of misclassified cases, providing deeper insight into the model's decision-making process beyond standard post-hoc explanations.
- The development of a fully offline, real-time desktop application that integrates classification and visual explanation, facilitating practical deployment in clinical environments with limited computational or network resources.
- Experimental evaluation demonstrating that a carefully designed lightweight CNN can outperform or match more complex models while offering improved interpretability and deployment readiness.

By bridging the gap between research innovation and clinical applicability, this work contributes a significant step forward in making AI-assisted brain tumor diagnosis more reliable and potentially useful for clinical decision support.

This paper is structured as follows: Section 2 reviews related work. Section 3 details the proposed methodology. Section 4 presents experimental results and clinical evaluation. Section 5 discusses key findings, and Section 6 concludes the paper with future directions.

2. Related work

This section reviews prior work on brain tumor classification, covering traditional machine learning methods, deep learning approaches, and recent explainable AI techniques used in medical imaging. From a clinical and imaging perspective, brain tumors present significant challenges in medical image analysis due to their varying types, locations, and sizes [9]. Brain tumors must be classified accurately to prepare a proper diagnosis and treatment plan. Researchers have employed different techniques to enhance the performance of classifiers, such as traditional machine learning and deep learning.

Traditional machine learning approaches have built on the foundation of handcrafted features and standard classifiers. Gumaie et al. [10] proposed a hybrid feature extraction technique combined with a Regularized Extreme Learning Machine (RELM), achieving 94.23% classification accuracy. Similarly, Kaplan et al. [11] utilized Local Binary Patterns (LBP and nLBP) with K-Nearest Neighbor (KNN) for classification, achieving 95.56% accuracy. Although these early approaches demonstrated promise, their dependence on manually crafted features restricted their scalability and ability to adapt to diverse datasets.

Deep learning methodologies, especially Convolutional Neural Networks (CNNs), have revolutionized medical image analysis and brain tumor detection due to their automated feature extraction and superior performance. CNNs excel in learning features directly from imaging data, bypassing the need for manual feature design. Afshar et al. [12] introduced a modified Capsule Network (CapsNet), emphasizing spatial relationships among features and achieving 90.89% accuracy. Rasheed et al. [13] proposed a CNN-based framework that incorporated Gaussian blur-based sharpening and Contrast-Limited Adaptive Histogram Equalization techniques, yielding a classification accuracy of 97.84% across four brain tumor classes. Furthermore, data augmentation techniques, such as scaling and rotation, have been incorporated to improve the performance of deep learning models [14].

Transfer learning has further advanced this field by enabling the reuse of powerful pre-trained models. Architectures such as VGG19, ResNet-50, InceptionV3, and DenseNet have been fine-tuned on domain-specific brain MRI datasets to achieve strong classification performance. Zhu et al. [15] demonstrated the effectiveness of transfer learning by fine-tuning the ResNet-18 architecture and selecting a randomized neural network as the classifier. Sharma et al. [16] modified ResNet-50 with additional custom layers. Mehnatkesh et al. [17] proposed a framework combining deep learning and evolutionary algorithms to optimize ResNet architecture, achieving an accuracy of 98.69%. Deepak and Ameer [18] proposed a brain tumor classification system using transfer learning with a pre-trained GoLeNet, achieving 98% accuracy on the Figshare MRI dataset using five-fold cross-validation.

While substantial progress has been achieved, challenges remain, particularly with respect to interpretability in CNN-based systems. To address this, several recent studies have integrated explainability techniques such as Grad-CAM to enhance transparency and facilitate clinical validation [19,20]. Li and Dib [9] applied Gradient-weighted Class Activation Mapping (Grad-CAM) to assess the interpretability of multiple pre-trained CNN models, including ResNet-50, Xception, and InceptionV3, enhancing transparency and facilitating clinical validation. Similarly, Musthafa et al. [21] proposed a brain tumor detection framework based on ResNet-50 integrated with Grad-CAM visualizations to support explainable decision-making.

Beyond CNN-based approaches, Vision Transformers have emerged as an alternative deep learning paradigm capable of capturing long-range dependencies within medical images [22]. Elhadidy et al. [23] compared CNNs, Swin Transformers, and EfficientNet for brain tumor classification, reporting that EfficientNet achieved the highest accuracy (98.72%) while maintaining favorable computational efficiency. Wang et al. [24] proposed a transformer-based architecture that combines token representations to enhance classification performance. Hybrid CNN-Transformer models have also been explored; for example, Aloraini et al. [25] combined convolutional feature extraction with self-attention mechanisms to capture both local and global representations. However, the increased computational and memory requirements of transformer-based models continue to pose challenges for routine clinical deployment. While these transformer-based and hybrid architectures can achieve competitive classification accuracy, their increased parameter counts (e.g., Swin Transformer-Tiny: 28.3M parameters) and higher inference latency pose challenges for deployment in resource-constrained clinical environments. Moreover, their explainability mechanisms often rely on post-hoc methods such as Grad-CAM, similar to our approach. The proposed lightweight CNN offers a more favorable trade-off between accuracy, computational efficiency, and interpretability, making it better suited for real-time clinical decision support. In parallel, recent studies have investigated federated and distributed learning paradigms to support multi-institutional collaboration while preserving data privacy, highlighting both opportunities and practical challenges for clinical translation of medical imaging AI systems [26,27].

Recent studies have explored optimization-assisted and robustness-oriented deep learning frameworks to improve convergence stability and generalization in complex medical imaging tasks. Hybrid bio-inspired optimization strategies and adaptive learning mechanisms have been reported to enhance training robustness and mitigate sensitivity to initialization and data variability, particularly in deep and transformer-based architectures [28–31]. More recently, uncertainty-aware and adaptive learning strategies have been proposed to improve robustness and interpretability under data variability, reinforcing their relevance to explainable AI in medical image analysis [32, 33]. While these approaches provide valuable insights into stability-aware learning, they often introduce additional architectural and optimization complexity, motivating the exploration of lightweight and deployment-oriented alternatives such as the framework proposed in this study.

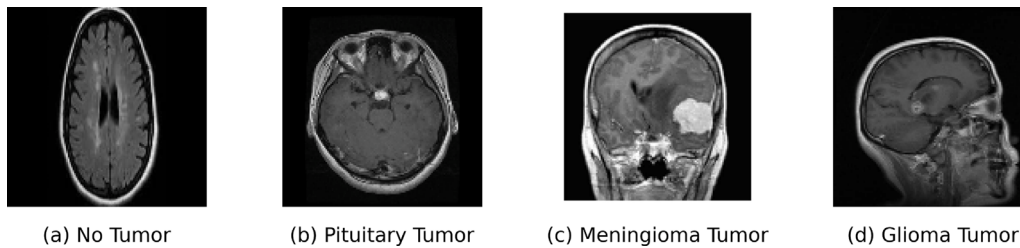


Fig. 1. Examples from the brain tumor MRI dataset, illustrating the variation in image appearance across the four classes.

3. Materials and methods

This section describes the dataset, preprocessing pipeline, model architecture, and explainability strategy adopted in this study. Each component is designed with consideration for classification accuracy, interpretability, and practical usability.

While recent studies have explored optimization-assisted and uncertainty-aware learning frameworks to enhance robustness, the proposed methodology emphasizes training stability and generalization through architectural design choices, regularization strategies, and explainable visualization, prioritizing computational efficiency and deployment simplicity.

3.1. Brain tumor MRI dataset

The dataset employed in this study is a publicly available image-level brain tumor MRI dataset obtained from Kaggle and curated by Kadam et al. [34]. It integrates MRI scans from several publicly accessible repositories, including Figshare, SARTAJ, and Br35H, thereby ensuring a diverse and representative collection of imaging data. It consists of 7023 MRI images, divided into four categories: normal (no tumor), pituitary tumors, meningioma tumors, and glioma tumors.

The dataset does not provide patient-level metadata (e.g., age, sex, clinical history, scanner manufacturer, magnetic field strength, or acquisition protocols), and subject identifiers are unavailable. Consequently, annotations are provided at the image level rather than the patient level. Labels were assigned by the dataset curators based on expert-reviewed sources and have been widely used in prior brain tumor classification studies. A sample of images from the dataset is shown in Fig. 1.

The dataset was systematically partitioned into training, validation, and testing subsets to ensure effective model development and evaluation. Specifically, 4569 images were allocated for training, 1143 images (20% of the training set) for validation, and 1311 images for testing.

This split preserves the overall dataset composition across the four classes: no tumor (2000 images, 28.5%), pituitary tumors (1757 images, 25.0%), meningioma tumors (1645 images, 23.4%), and glioma tumors (1621 images, 23.1%). The dataset exhibits mild class imbalance, with the ‘no tumor’ class exceeding the smallest class (glioma) by approximately 5.4 percentage points. To mitigate potential bias, stratified splitting was employed to maintain class proportions across all subsets. Macro-averaged evaluation metrics were used to ensure balanced performance assessment, and data augmentation was applied uniformly to the training set without introducing artificial rebalancing techniques such as oversampling or class weighting. The class distribution and its preservation across splits are illustrated in Fig. 2.

3.1.1. Data preprocessing

Effective image preprocessing plays a vital role in enhancing the performance of deep learning models, especially in medical imaging applications. To get the dataset ready for training, it is subjected to various preprocessing steps. All images are resized to 125×125 pixels in the first step. In the next step, pixel values are normalized by dividing them by 255. This normalization step helps stabilize training

and improves optimization efficiency. The training data were randomly shuffled to reduce sampling bias and improve generalization. Class labels were one-hot encoded, which is required for multi-class classification.

3.1.2. Data augmentation

The specialized nature of medical imaging datasets gives them high value but also comes with several challenges. Accurate annotation requires domain-specific expertise, while labeling inconsistencies can result in negatively influence model training and predictive performance [35]. In addition, when it comes to data privacy, the sensitive nature of the data limits the accessibility and sharing of data. As a result, it complicates the data gathering of sufficiently large and representative datasets.

The limited size of labeled medical image datasets creates a severe bottleneck to effective deep learning model development. To avoid this problem, the image augmentation techniques have been widely used to increase data diversity to improve model generalization synthetically [36]. Consistent with prior brain tumor classification studies, data augmentation has been extensively employed to mitigate overfitting and enhance model robustness when training on publicly available MRI datasets. Commonly adopted strategies in the literature include geometric transformations such as rotation, flipping, and translation [37,38], while more advanced approaches have explored GAN-based synthesis to generate realistic tumor variations [39,40]. Compared to these methods, our augmentation strategy prioritizes computational efficiency and anatomical preservation. In this study, data augmentation was employed to enhance robustness and improve generalization under real-world variability. The augmentation strategy included controlled geometric transformations with the following parameters: rotation range = 10, width shift range = 0.05, height shift range = 0.05, and horizontal flipping. The rotation range introduces random rotations up to ± 10 degrees, while the shift parameters apply minor horizontal and vertical translations of up to 5% of the image dimensions. Horizontal flipping was applied with a probability of 0.5, further increasing sample diversity. All augmentation operations were applied exclusively to the training set, while validation and test data were kept unchanged to ensure unbiased performance evaluation.

Fig. 3 illustrates the impact of data augmentation by presenting a visual comparison between original and augmented MRI scans. The top row displays four representative training images in their original form, while the bottom row depicts the corresponding images following the application of augmentation transformations.

3.2. Proposed model architecture

Convolutional Neural Networks (CNNs) have emerged as a powerful approach for automatic feature extraction, achieving state-of-the-art performance in various image classification tasks. Their ability to identify complex image patterns makes them particularly effective for medical image analysis [41].

In this work, a custom lightweight CNN architecture is adopted. This design provides explicit control over feature hierarchies and activation behavior, reduces computational and memory overhead, and supports

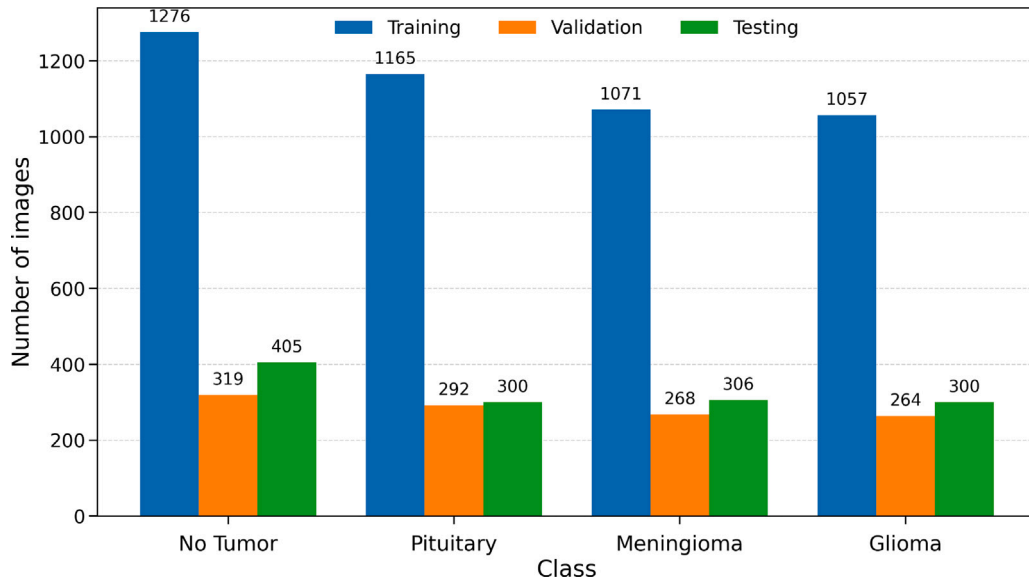


Fig. 2. Class-wise distribution of MRI images across tumor categories in the training, validation, and testing splits.

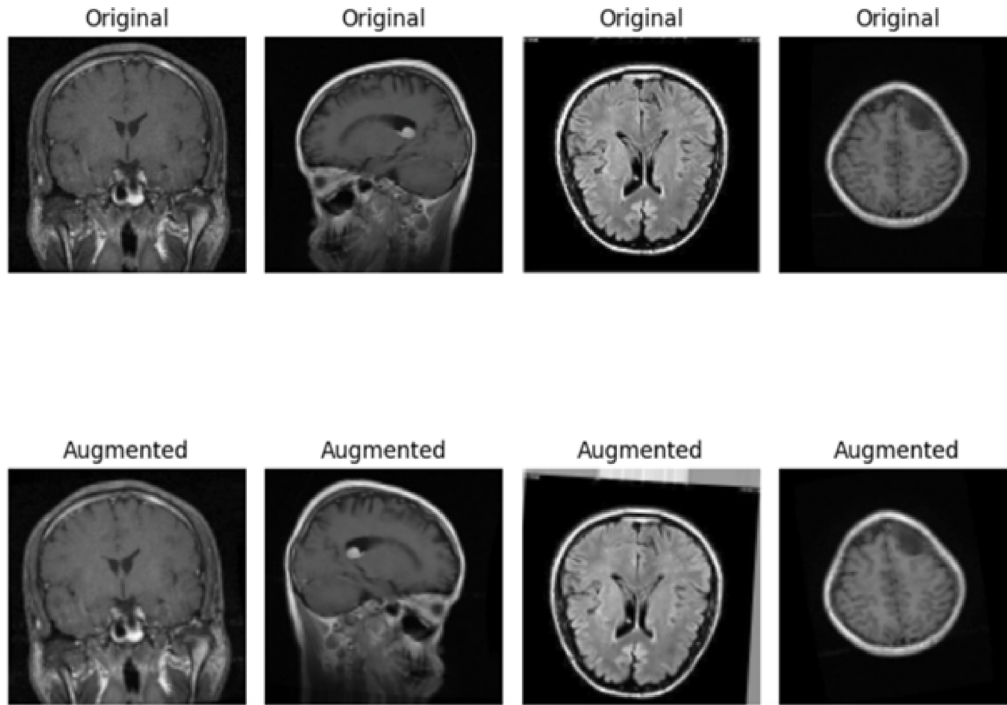


Fig. 3. Original (top) and augmented (bottom) MRI samples illustrating the effect of data augmentation.

effective gradient-based explainability capabilities that are more difficult to achieve when relying on deeper transfer learning or transformer-based backbones. These characteristics are important for real-time decision-support scenarios in clinical settings and for generating reliable, interpretable visual explanations via Grad-CAM. The proposed design aligns with recent adaptive and uncertainty-aware learning principles [32,33,42], incorporating batch normalization, dropout, and data augmentation to enhance stability and generalization without complex optimization overhead.

The proposed approach employs a custom-designed Convolutional Neural Network (CNN) architecture tailored for brain tumor classification, as depicted in Fig. 4. Each convolutional block within the model comprises a Conv2D layer with a 3×3 kernel, followed sequentially

by batch normalization and a ReLU activation function to ensure stable and non-linear feature transformation.

The model architecture begins with an input layer that specifies the dimensionality of the input data without performing any feature extraction. In this study, the input consists of grayscale MRI images with a fixed resolution of 125×125 pixels. The images are then passed through a number of hidden layers that learn to represent features in a hierarchy to facilitate correct classification. The feature extraction process consists of five convolutional blocks. The first block consists of two Conv2D layers with 64 filters each, followed by batch normalization. Further, a max-pooling layer reduces the spatial dimensions of the input. Finally, a dropout layer of 25% reduces overfitting. The second block consists of two 128-filter Conv2D layers. They also have

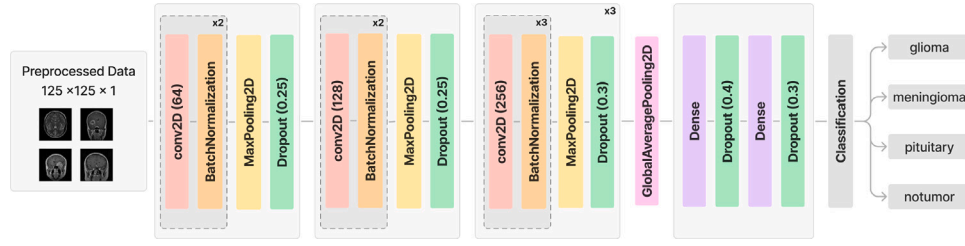


Fig. 4. Architecture of the proposed lightweight CNN model.

batch normalization, max-pooling, and a 25% dropout layer. The third, fourth, and fifth blocks include three Conv2D layers at each block with 256 filters, followed by batch normalization and max-pooling, and a dropout of 30% was used at the deeper level for regularization.

The convolutional layers take complex features from the images [43, 44]. The training stability is enhanced through batch normalization, while max-pooling decreases spatial dimensionality and computational cost by emphasizing the most salient features [45]. Dropout layers were employed to reduce overfitting by randomly deactivating neurons during training.

Following the convolutional feature extraction stages, the model adds a GlobalAveragePooling2D layer that reduces the size of the feature maps, which also turns 2D representations into a 1D vector for classification purposes. This gets the resulting high-dimensional feature maps ready for the fully connected layers.

After feature extraction, the feature maps are passed through the fully connected layers for final classification. The first dense layer comprises 256 units with ReLU activation, followed by a 40% dropout layer to mitigate overfitting. A second dense layer of 64 units with ReLU activation is then applied, followed by a 30% dropout layer. Finally, the output layer consists of four units corresponding to the four tumor classes, employing a softmax activation function to produce class probabilities.

3.3. Visualization of model decision-making with Grad-CAM

Providing insights into the internal reasoning of the CNN model is critical for building clinical trust. Unlike performance-driven evaluation, explainability analysis focuses on whether a model attends to clinically meaningful regions, offering validation beyond numerical metrics [46].

Gradient-weighted Class Activation Mapping (Grad-CAM) generates class-discriminative visualizations by leveraging gradients flowing into the final convolutional layer [46]. For class c , importance weights α_k^c for each feature map A^k are computed via global average pooling of gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (1)$$

The Grad-CAM heatmap is then obtained as a weighted combination of feature maps followed by ReLU:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (2)$$

ReLU ensures only features positively influencing the target class are highlighted. The final convolutional layer is chosen because it preserves spatial information while encoding high-level semantics.

In this study, Grad-CAM visualizations verify that predictions are based on clinically relevant tumor regions rather than artifacts [47], enabling clinicians to cross-validate model decisions and supporting trustworthy diagnostic aids.

Table 1

Model training configuration and hyperparameters.

Component	Setting/Value
Optimizer	Adamax
Learning rate	0.001 (fixed; no scheduling)
Batch size	8
Epochs	50
Loss function	Categorical Cross-Entropy
Output activation	Softmax
Dropout rates	0.25, 0.3, 0.4
Early stopping	Not applied
Model checkpointing	Best validation epoch
Data augmentation	Training set only

4. Experimental evaluation and clinical readiness of the proposed framework

This section presents the experimental setup, evaluation metrics, and quantitative and qualitative results. The analysis focuses on classification performance, generalization behavior, and interpretability through Grad-CAM visualizations.

4.1. Experimental setup and training configuration

The model was trained using GPU acceleration with TensorFlow/Keras to improve computational efficiency. Google Colab Pro+ was used in this study, as it provides access to high-performance GPUs, extended session durations, and increased memory resources. Specifically, NVIDIA T4 and A100 GPUs were utilized to support efficient training of deep learning models.

4.1.1. Training configuration

The model's hyperparameters were selected to balance classification performance, generalization ability, and computational efficiency. Hyperparameters were selected empirically through iterative experimentation, where candidate configurations (learning rate, batch size, dropout rates, and optimizer) were evaluated based on validation performance and training stability. The reported settings represent the best-performing combination among the tested configurations. The final training configuration, chosen to facilitate exact replication, is summarized in Table 1. The Adamax optimizer was selected for its adaptive learning rate behavior in complex classification tasks [8]. The dropout rates (0.25, 0.3, 0.4) and the 50-epoch training limit were empirically validated to mitigate overfitting. A batch size of 8 ensured stable training given the dataset size. Data augmentation was applied exclusively to the training set to maintain unbiased evaluation on the validation and test splits. All hyperparameters were kept fixed across the cross-validation experiments and ablation studies to ensure a fair and reproducible evaluation of the proposed architecture. Using this fixed configuration, the final model was trained for a total duration of approximately 32.5 min (1948 s) on an NVIDIA T4 GPU, with an average training time of approximately 39 s per epoch.

Table 2

Stratified 5-fold cross-validation performance of the proposed framework. Results are reported as mean $\pm$ standard deviation with 95% confidence intervals (CI).

Metric	Mean $\pm$ SD (%)	Range (%)	95% CI (%)
Accuracy	99.29 $\pm$ 0.39	[98.65–99.72]	[98.81–99.77]
AUC	99.92 $\pm$ 0.05	[99.86–99.98]	[99.86–99.98]
F1-score	99.29 $\pm$ 0.39	[98.65–99.72]	[98.81–99.77]

4.1.2. Performance metrics and quantitative evaluation

The model's performance in brain tumor classification was evaluated using multiple quantitative metrics. Confusion matrices were employed to analyze prediction outcomes, including true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). Based on these values, standard classification metrics—accuracy, precision, recall, and F1-score were computed to assess overall performance [48].

- **Accuracy:** quantifies the proportion of correct predictions out of all predictions made by the classifier. It is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

- **Precision:** measures the proportion of correctly identified positive predictions out of all predicted positives:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

- **Recall:** (or sensitivity) indicates the proportion of actual positive cases that were correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

- **F1-score:** represents the harmonic mean of precision and recall, offering a balance between the two:

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

4.2. Results and analysis

This section evaluates the proposed method using standard metrics, visual analyses, and comparisons with state-of-the-art approaches. It also explores the model's potential for real-world deployment in clinical workflows.

4.2.1. Quantitative analysis

The proposed brain tumor classification system was comprehensively evaluated using quantitative performance metrics and analysis of training behavior. The model achieved a test accuracy of 99.54%, demonstrating strong classification capability across the four tumor classes. The precision, recall, and F1-score on the test set were also 99.54%, indicating consistent predictive performance with low false positive and false negative rates.

To further assess robustness and mitigate potential sensitivity to data partitioning, stratified 5-fold cross-validation was performed. Across folds, the proposed framework achieved a mean accuracy of 99.29% $\pm$ 0.39% with a narrow 95% confidence interval, alongside a consistently high AUC of 99.92% $\pm$ 0.05%. These results indicate stable performance across multiple data splits and suggest that the reported accuracy is not dependent on a single train–test configuration (see Table 2).

The confusion matrix (Fig. 5) illustrates the classification performance of the proposed model on the held-out test set across the four tumor categories. *No tumor* cases were classified with 100% accuracy, while *pituitary*, *meningioma*, and *glioma* achieved recalls of 99.7%, 99.0%, and 99.0%, respectively. To ensure a balanced assessment in the presence of potential class imbalance, per-class precision, recall, and F1-scores obtained on the held-out test set are reported in Table

Table 3

Per-class classification performance on the held-out test set.

Class	Precision (%)	Recall (%)	F1-score (%)
no tumor	100.00	100.00	100.00
pituitary	100.00	99.67	99.83
meningioma	98.70	99.02	98.86
glioma	99.00	99.00	99.00

Table 4

Model complexity and efficiency comparison. Parameter counts are reported from the literature, while GFLOPs and inference time are measured under the same experimental setup.

Model (Architecture type)	Params (M)	GFLOPs	Inference (ms)
Standard & Attention CNNs			
ResNet-50	25.6	7.80	4.23 $\pm$ 1.28
SE-ResNet-50	28.1	7.75	5.12 $\pm$ 1.38
Transformer-based			
ViT-B/16	86.6	33.81	15.90 $\pm$ 0.13
Swin-Tiny	28.3	8.74	5.83 $\pm$ 0.24
Hybrid			
CvT-13	20.0	8.15	5.72 $\pm$ 0.23
Proposed lightweight CNN	5.37	0.4288	0.66 $\pm$ 0.11

3. Misclassifications were minimal and primarily occurred between pituitary tumors and either meningioma or glioma cases, likely due to overlapping morphological features in MRI scans. These low error rates (approximately 0.3%–1.6%) indicate robust and consistent classification behavior. This performance is supported by the use of stratified data splitting to preserve class distributions and targeted data augmentation applied to the training set to enhance sample diversity.

Precision–recall (PR) curve analysis was performed using a one-vs-rest strategy to evaluate class-wise performance under varying decision thresholds. As shown in Fig. 6, the proposed framework achieves near-perfect average precision across all classes, with AP values of 1.000 for *no tumor*, *pituitary*, and *glioma*, and 0.999 for *meningioma*. These results indicate strong robustness to class imbalance and reliable performance at high recall levels.

The training and validation curves shown in Fig. 7 further confirm the model's stable learning behavior and generalization capability. Accuracy improves steadily with a minimal gap between training and validation curves, while the loss decreases consistently and stabilizes after approximately 20 epochs. This close alignment indicates limited overfitting and reflects the effectiveness of the adopted preprocessing, data augmentation, and regularization strategies.

Table 4 presents a quantitative comparison of model complexity between the proposed framework and representative state-of-the-art architectures, including standard CNNs, attention-based CNNs, vision transformers, and hybrid convolution–transformer models. While parameter counts are reported from the literature for consistency, GFLOPs and inference time were measured under the same experimental setup to ensure a fair and controlled comparison. In addition to its compact design (5.37M parameters), the proposed model achieved an average inference latency of 0.66 $\pm$ 0.11 ms per image under the experimental hardware configuration (NVIDIA T4 GPU). The proposed lightweight CNN demonstrates significantly lower computational complexity, requiring only 0.4288 GFLOPs, compared to transformer-based models such as ViT-B/16, which incur substantially higher computational cost and slower inference. Even more efficient transformer variants, such as Swin Transformer-Tiny and CvT-13, remain considerably heavier than the proposed model. These results highlight the lightweight nature of the framework and its suitability for computationally constrained and time-sensitive medical image analysis applications.

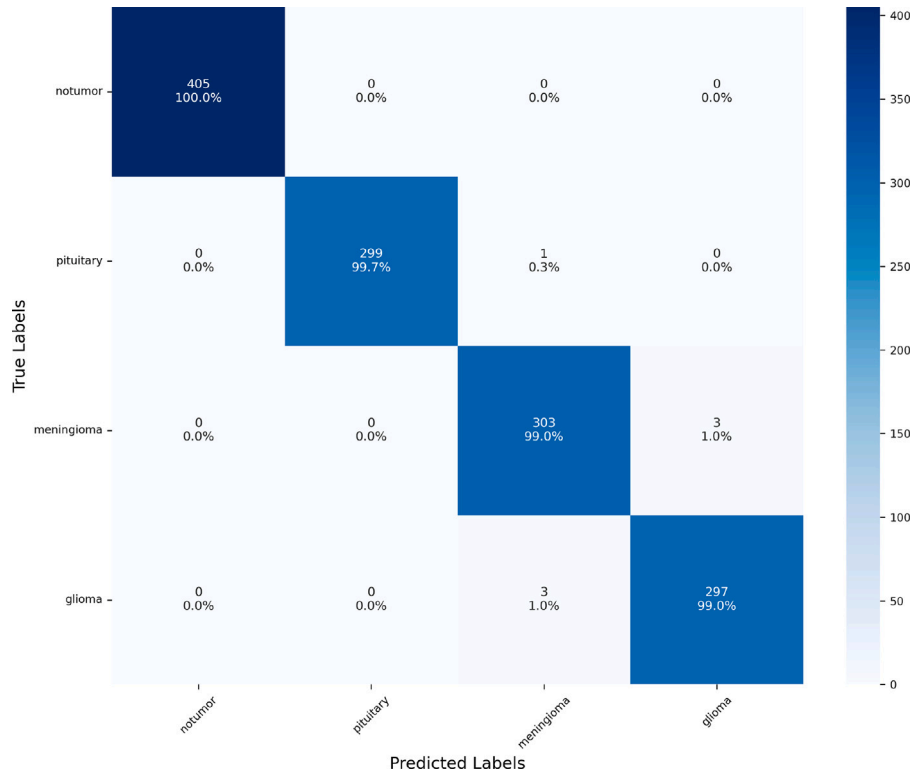


Fig. 5. Confusion matrix showing classification performance on the test set.

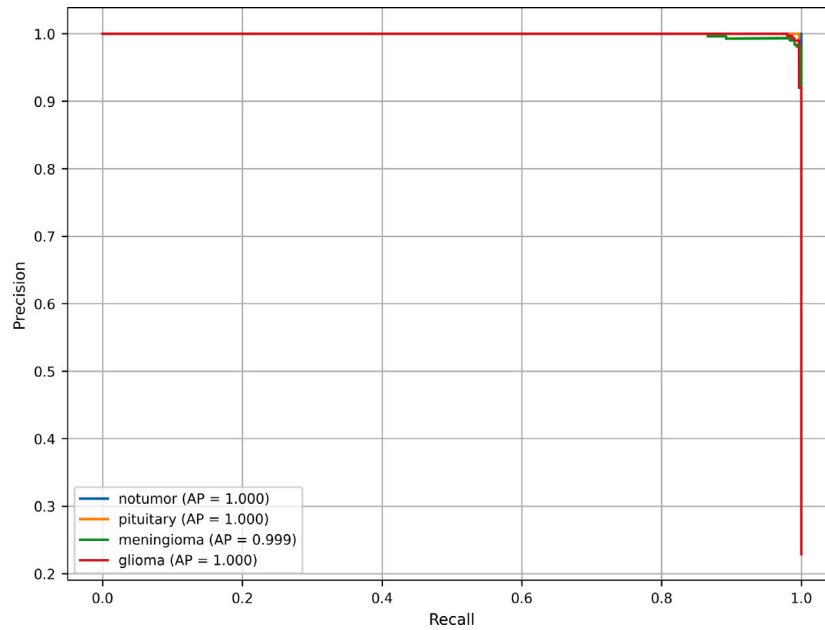


Fig. 6. Precision-recall curves (one-vs-rest) for each tumor class.

4.2.2. Ablation study

An ablation study was conducted to systematically evaluate the contribution of key architectural and training components while keeping the training protocol, data splits, and optimization settings unchanged. Each ablation configuration was trained from scratch using identical hyperparameters to ensure fair comparison. The evaluated factors included network depth, batch normalization, dropout regularization strength, and data augmentation. Table 5 summarizes the performance variations using macro-averaged metrics.

The results confirm that all components contribute to overall performance. Removing data augmentation (A1) leads to a moderate decline (F1-score: -1.01 percentage points), while eliminating batch normalization (A2) causes the largest degradation (F1-score: -5.01 percentage points), highlighting its critical role in training stability. Reducing network depth (A3) by removing the deeper convolutional blocks and limiting hierarchical feature extraction results in a notable drop (F1-score: -3.17 percentage points), underscoring the importance of deeper semantic feature representation for accurate tumor discrim-

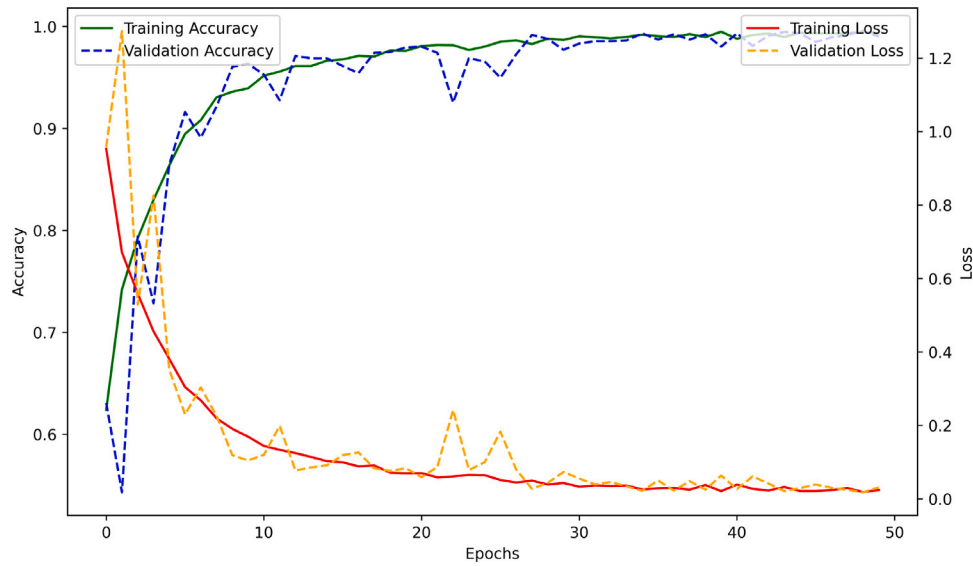


Fig. 7. Training and validation accuracy and loss over 50 epochs.



Fig. 8. Representative correctly classified MRI samples with true and predicted labels (green indicates correct predictions).

Table 5

Ablation study evaluating the contribution of individual framework components. Performance is reported using macro-averaged metrics (%) on the test set.

ID	Configuration	Acc.	Prec.	Rec.	F1	$\Delta F1^a$
A0	Full proposed framework	99.54	99.54	99.54	99.54	–
A1	Without data augmentation	98.63	98.55	98.51	98.53	–1.01
A2	Without batch normalization	94.74	94.83	94.36	94.53	–5.01
A3	Shallower network	96.49	96.51	96.28	96.37	–3.17
A4	Reduced regularization	98.47	98.43	98.35	98.39	–1.15

^a Change in macro F1-score relative to the full model (A0), expressed in percentage points.

ination. Lowering dropout regularization strength (A4) also decreases performance (F1-score: –1.15 percentage points), indicating the benefit of stronger regularization in the full model.

4.2.3. Visualization of correctly classified MRI samples

To complement the quantitative evaluation, Fig. 8 presents representative MRI scans alongside their true and predicted labels. The model correctly classifies samples across all tumor categories, including *meningioma*, *pituitary*, *glioma*, and *no tumor*, with correct predictions indicated by green-highlighted labels. These qualitative results are consistent with the reported quantitative metrics and provide visual confirmation of the model's classification behavior.

4.2.4. Visualization of model decision-making using Grad-CAM

To examine model interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) was employed to visualize regions of the input MRI scans that most influenced the model's predictions. Grad-CAM

produces class-discriminative heatmaps that highlight salient image regions without requiring architectural modifications. Fig. 9 illustrates the evolution of model attention across multiple convolutional layers for a correctly classified *meningioma* sample. Early layers exhibit diffuse responses to general anatomical structures, while deeper layers progressively focus on the tumor region. This coarse-to-fine localization behavior is consistent with the hierarchical feature learning characteristics of convolutional neural networks.

To further assess model explainability, Figs. 10 and 11 present Grad-CAM overlays for both correctly classified and misclassified samples across the four tumor classes: *glioma*, *meningioma*, *pituitary*, and *no tumor*. The top row in each figure illustrates accurate predictions, where attention heatmaps are well aligned with the tumor mass. In contrast, the bottom row highlights misclassified cases, revealing that the model occasionally focuses on irrelevant anatomical structures (e.g., ventricles or image borders), which may contribute to confusion between tumor types. For instance, in Fig. 10, several *no tumor* cases are misclassified as *glioma*, with Grad-CAM activations concentrated near artifact regions or ambiguous tissue boundaries. Similarly, Fig. 11 shows that *pituitary* tumors are misinterpreted as *glioma* or *meningioma* when the model's attention drifts toward peripheral features rather than the central lesion. These observations demonstrate the utility of Grad-CAM not only for validating correct predictions but also for diagnosing common failure modes and potential model biases.

A layer-wise Grad-CAM analysis was further conducted on representative correct and misclassified cases (Fig. 12). In the correctly classified example, deeper-layer activations accurately localize the tumor region, while earlier layers exhibit broader responses. In contrast, misclassified cases show diffuse or misplaced activations and insufficient focus on diagnostically relevant regions at deeper layers. For

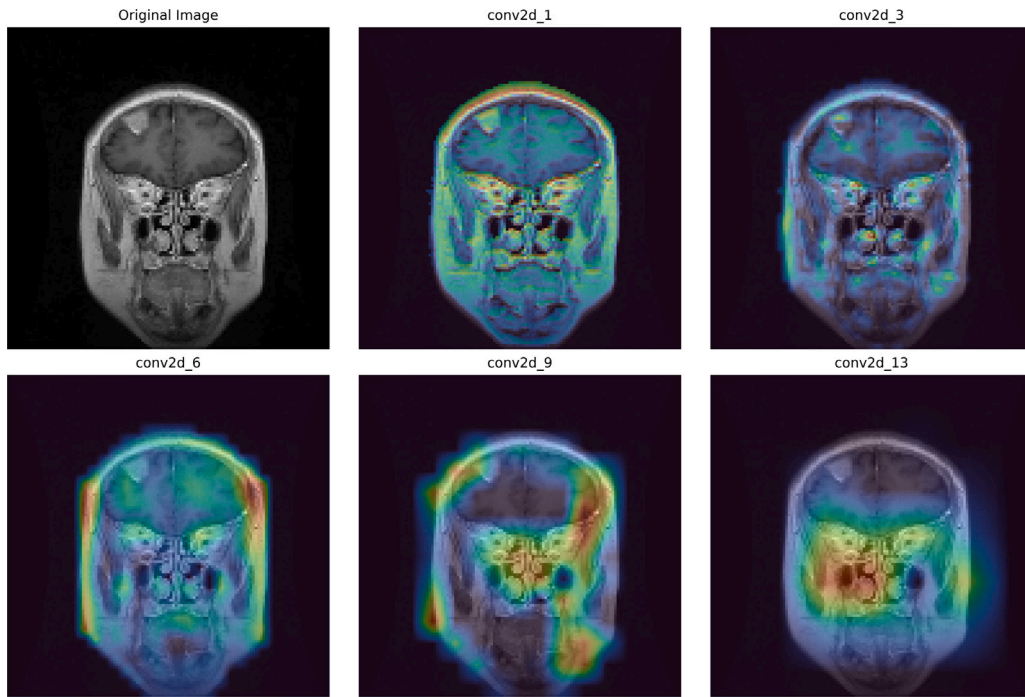


Fig. 9. Layer-wise Grad-CAM attention maps for a correctly classified meningioma sample.

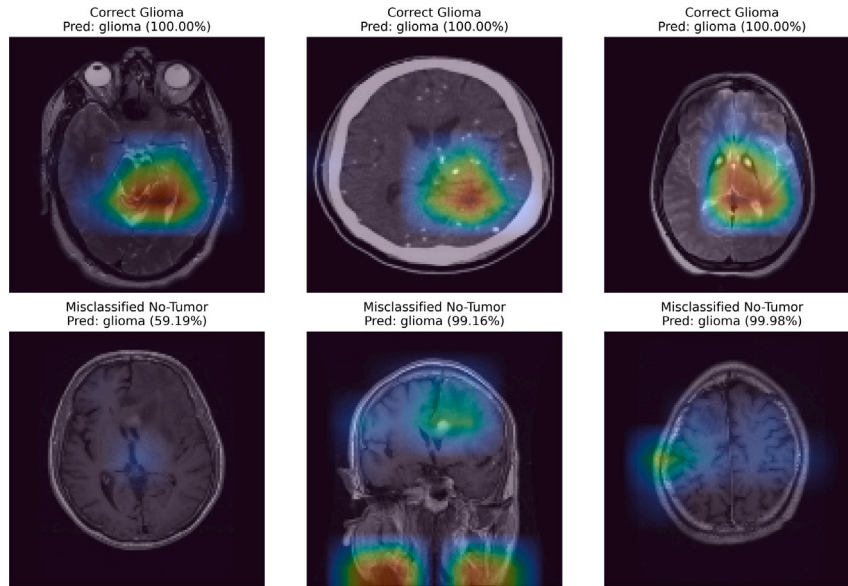


Fig. 10. Correctly classified glioma (top) and misclassified no tumor cases (bottom) with Grad-CAM overlays.

misclassified samples, Grad-CAM was computed with respect to the ground-truth class to assess whether salient tumor features were present but under-emphasized by the model.

The Grad-CAM visualizations in Figs. 9–12 qualitatively illustrate that the model's attention is primarily concentrated within the anatomical brain region. To verify this observation quantitatively, we evaluate the spatial consistency of the Grad-CAM activations using two localization metrics: the Attention Focus Score (AFS) and the Peak-in-Brain Ratio (PBR). To complement the qualitative Grad-CAM visualizations, we perform a quantitative evaluation of attention localization. Recent work has emphasized that qualitative assessments alone are insufficient for establishing clinical trust in saliency methods, motivating the need for rigorous quantitative evaluation [49,50]. However, since our dataset does not provide pixel-level tumor segmentation masks,

overlap-based metrics such as Dice or Intersection-over-Union cannot be computed directly, as they require pixel-level annotations [51]. We therefore introduce two metrics tailored to brain MRI analysis that leverage anatomical structure instead of requiring tumor annotations. The **Attention Focus Score (AFS)** measures the fraction of total Grad-CAM activation energy concentrated within the anatomical brain region B :

$$\text{AFS} = \frac{\sum_{(i,j) \in B} H_{i,j}}{\sum_{(i,j)} H_{i,j}}, \quad (7)$$

where $H_{i,j}$ is the heatmap intensity at pixel (i, j) and B represents the binary brain mask derived from the MRI slice. The **Peak-in-Brain Ratio (PBR)** reports the proportion of samples where the maximum Grad-CAM response lies inside the brain mask. These metrics adapt

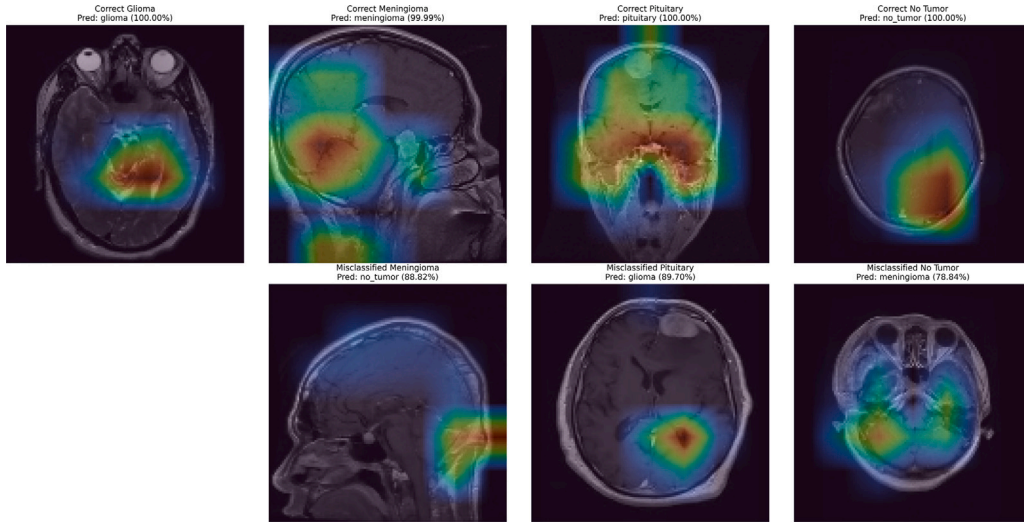


Fig. 11. Representative correct and misclassified cases with Grad-CAM visualizations for each tumor class.

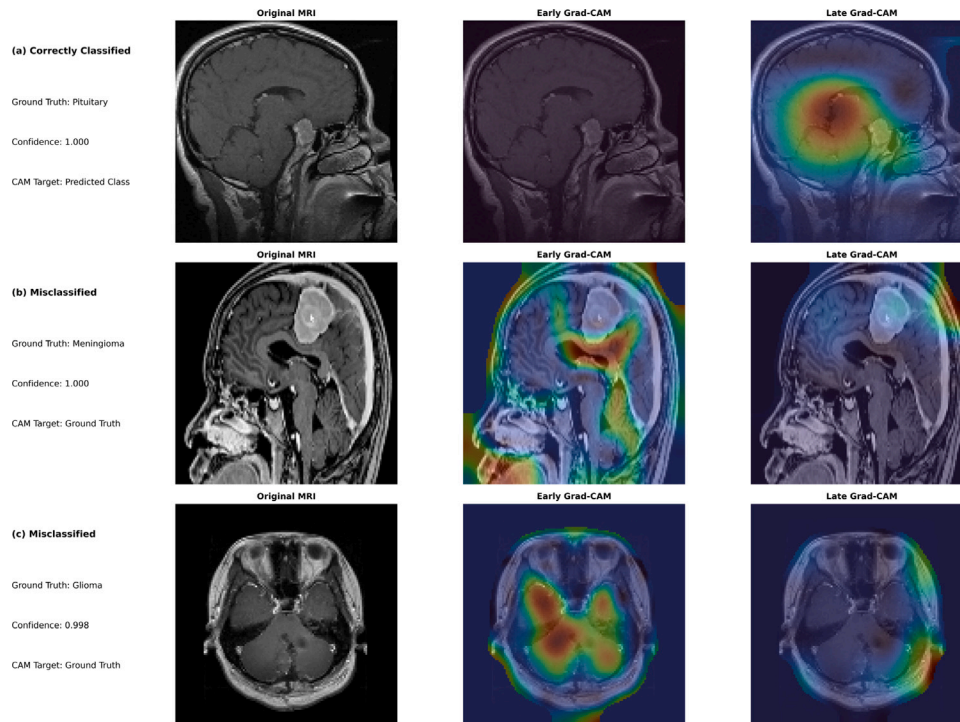


Fig. 12. Layer-wise Grad-CAM analysis for representative correct and misclassified cases.

the principle of localization evaluation [49] to a setting where only anatomical region annotations (i.e., brain masks) are available. Across the 1293 correctly classified test samples, the model achieved a mean AFS of 0.957 ± 0.042 and a PBR of 0.972, indicating strong spatial focus within brain tissue. These metrics provide a quantitative proxy for localization performance in the absence of pixel-level tumor annotations, thereby strengthening the validation of Grad-CAM explanations beyond qualitative visualization. These results demonstrate that approximately 95.7% of the model's explanatory focus is confined to brain tissue, with peak activations located within the brain in over 97% of cases. This provides quantitative evidence that the model's decisions are predominantly guided by anatomically meaningful regions.

The Attention Focus Score (AFS) and Peak-in-Brain Ratio (PBR) provide insight into whether the model's attention is anatomically consistent, which is a necessary, although not sufficient, condition for

trustworthy predictions. However, standard localization metrics such as Dice or Intersection over Union (IoU) cannot be computed in this study due to the absence of pixel-level tumor annotations. Consequently, unlike Dice or IoU, the proposed metrics do not quantify overlap with precise tumor boundaries, but instead evaluate attention at the organ level (brain) rather than at the lesion level. These metrics are therefore not intended to replace segmentation-based evaluation, but to provide complementary insight under weakly supervised conditions. Future work incorporating tumor segmentation datasets will enable direct comparison with Dice/IoU and a more fine-grained assessment of localization accuracy.

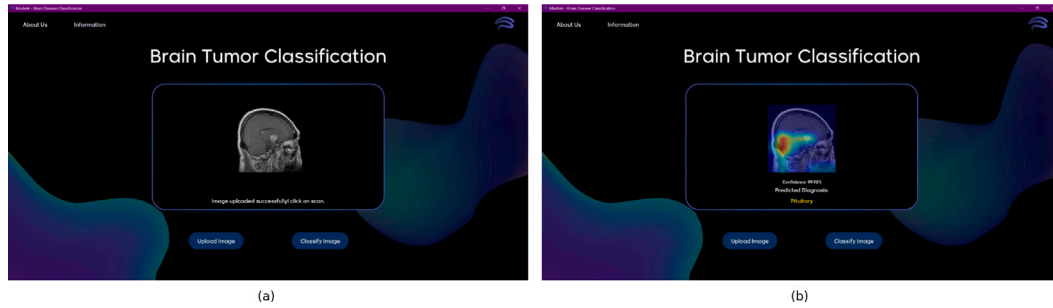
4.3. Comparison with state-of-the-art methods

To further validate the performance of the proposed model, a detailed comparative analysis with recent state-of-the-art brain tumor

Table 6

Comparison of the proposed model with recent brain tumor classification approaches based on methodology, number of classes, and evaluation metrics.

Authors	Methodology	Classes	Precision	Recall	F1-score	Accuracy
Özkaraca et al. [52]	VGG16 + DenseNet	4	96.00%	96.50%	96.00%	97%
Gómez-Guzmán et al. [53]	Pre-trained CNNs	4	97.97%	96.59%	97.27%	97.12%
Raouf et al. [54]	DCST + SVM	2	97.80%	96.60%	97.20%	97.71%
Rasheed et al. [13]	Image enhancement + CNN	4	97.85%	97.85%	97.90%	97.84%
Alnemr et al. [55]	Modified ResNet152V2	4	—	—	—	98.90%
Li et al. [9]	ResNet-50	4	99.00%	99.00%	99.00%	98.70%
Proposed model	Custom CNN	4	99.54%	99.54%	99.54%	99.54%

**Fig. 13.** Real-time desktop application interface: (a) MRI upload; (b) classification result with Grad-CAM overlay.

classification methods is presented in Table 6. The comparison includes diverse methodologies: dense convolutional networks, transfer learning, hybrid feature extraction, and SVM classifiers. Evaluation metrics including precision, recall, F1-score, and overall classification accuracy are reported across studies addressing both binary and multi-class (four-class) classification tasks.

As shown, the proposed model consistently outperforms prior work, achieving 99.54% across all metrics (precision, recall, F1-score, accuracy). These results demonstrate strong generalization and clinical suitability. However, direct comparisons should be interpreted cautiously due to differences in datasets, splits, and evaluation protocols.

4.4. Real-time brain tumor detection application

To demonstrate the practical utility of the proposed model, we developed a user-friendly desktop application using Python, Tkinter, and CustomTkinter for real-time brain tumor detection. The application allows healthcare professionals to upload MRI scans, initiate model inference, and receive classification results instantly—streamlining the diagnostic workflow.

The system supports classification into four categories: *no tumor*, *pituitary*, *meningioma*, and *glioma*. It provides both the predicted class and the associated confidence score. To enhance interpretability, Grad-CAM visualizations are incorporated to highlight the image regions most influential in the model's decision.

Fig. 13 illustrates the application interface. In Fig. 13a, the user has uploaded an MRI scan and is prompted to classify it. Upon clicking the “Classify Image” button, the application displays the predicted tumor type (e.g., *pituitary*), the confidence score (e.g., 99.98%), and a Grad-CAM overlay highlighting the tumor region (Fig. 13b).

To support interpretability and streamline clinical workflows, the interface employs a color-coded scheme—green for *no tumor*, yellow for *pituitary*, orange for *meningioma*, and red for *glioma*—allowing rapid visual assessment. The design prioritizes simplicity and accessibility, ensuring usability even for non-technical users. Additionally, the system operates entirely offline, making it suitable for clinics with limited internet connectivity or heightened data security requirements.

This application functions as a real-time decision support tool, helping radiologists review cases efficiently and confidently, particularly in resource-constrained or high-demand clinical settings.

5. Discussion

This section interprets the results with emphasis on clinical relevance, deployment considerations, and study limitations.

5.1. Interpretation and advantages

The proposed framework achieved a classification accuracy of 99.54% while maintaining a lightweight design (5.37M parameters), enabling real-time inference at approximately 0.66 ms per image (pure model forward pass). Beyond accuracy, Grad-CAM visualizations provided clinically meaningful validation through anatomically plausible attention patterns, addressing the critical need for transparency and interpretability in medical AI systems. Together, these results demonstrate a strong balance between performance, efficiency, and explainability, representing a practical step toward clinical integration.

5.2. Clinical translation

The framework is designed as an offline decision-support tool, supporting data privacy and usability in resource-limited clinical environments. Its integrated explainability mechanisms facilitate clinician trust and align with emerging regulatory expectations for transparent AI-assisted diagnostics.

In real-world clinical settings, MRI data exhibit substantial variability due to differences in scanner manufacturers, magnetic field strengths (e.g., 1.5T vs. 3T), and acquisition protocols (e.g., T1, T2, FLAIR sequences). In addition, patient-specific factors such as age, tumor characteristics, and anatomical variability further contribute to data diversity. These factors can introduce domain shifts that may affect model performance when deployed outside controlled experimental conditions. For example, a model trained predominantly on T1-weighted contrast-enhanced images may exhibit reduced sensitivity when applied to non-contrast T2-weighted sequences. Furthermore, real-world clinical populations may exhibit class distributions that differ from those in benchmark datasets, which may influence model calibration and decision thresholds.

Future multicenter validation and prospective clinical studies are essential to address real-world challenges such as scanner variability, protocol heterogeneity, and patient-level diversity, and to support regulatory approval and clinical adoption.

We also acknowledge that retrospective dataset evaluation cannot fully replicate prospective clinical deployment, where factors such as image quality variations, acquisition timing, and patient condition introduce additional uncontrolled sources of variation.

5.3. Limitations and future work

A key limitation of this study is the reliance on a single publicly available dataset, which may not fully capture the diversity and variability encountered in real-world clinical environments. In particular, the dataset is curated at the image level and lacks subject-level identifiers and detailed acquisition metadata (e.g., scanner type, imaging protocols, and patient demographics), which prevents direct assessment of cross-institutional generalization.

Although the proposed framework demonstrates strong and consistent performance, including under stratified cross-validation, this evaluation remains confined to a single data distribution. Consequently, performance may vary when the model is applied to external datasets with different imaging characteristics and acquisition conditions. Given the high classification accuracy achieved, particular care was taken to minimize the risk of overfitting and data leakage. The model exhibited consistent performance across cross-validation folds, suggesting stable learning behavior rather than overfitting to a specific data split.

To mitigate this limitation, strict experimental protocols were adopted, including complete separation of training, validation, and test sets, stratified cross-validation, and controlled data augmentation to simulate real-world variability. Nevertheless, these strategies cannot fully substitute for true external validation. The held-out test set was fully isolated and was not used during preprocessing, augmentation, or hyperparameter tuning. However, due to the absence of subject-level identifiers, potential hidden correlations cannot be entirely ruled out and should be considered when interpreting the results.

Future work will therefore focus on external validation using independent and multi-institutional MRI datasets to assess robustness under domain shifts. In addition, federated learning and domain adaptation strategies will be explored to enable generalization across heterogeneous clinical settings while preserving data privacy.

Furthermore, robustness-enhancing strategies such as class-weighted loss functions and imbalance-aware training schemes will be investigated to improve performance on rare tumor subtypes and highly imbalanced data. Future work may also investigate the integration of uncertainty-aware or adaptive optimization strategies to further enhance robustness and interpretability under increased data variability, as highlighted in recent studies [32,33,42].

Beyond qualitative Grad-CAM visualizations, recent studies have proposed quantitative metrics to evaluate explainability in medical imaging, including fidelity (insertion/deletion scores), stability (consistency under perturbations), localization (Dice, IoU, Pearson correlation), region-based scoring, and computational efficiency [56,57]. Incorporating such metrics in future work would complement the current qualitative analysis and enable a more rigorous and clinically meaningful assessment of model interpretability.

6. Conclusion

This study presents an interpretable deep learning framework for multi-class brain tumor classification from MRI scans. The proposed model achieves high classification accuracy while maintaining computational efficiency suitable for practical clinical decision-support scenarios. Multi-level Grad-CAM analysis provides clinically meaningful transparency, while the fully offline desktop implementation supports potential use in resource-limited settings. While results under stratified cross-validation demonstrate robust performance, external multicenter validation remains necessary to further assess generalization across diverse clinical imaging environments.

CRedit authorship contribution statement

Helala AlShehri: Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Luluh Almogbil:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Conceptualization. **Jiyan Alrajhi:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Lama Alo-saimi:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Jawlan Almarri:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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